

Solved: DBMS NoSQL Assignment

Comprehensive Solutions & MongoDB Queries

Section A: Theory Questions

1. Define NoSQL databases. How do they differ from traditional relational databases?

NoSQL (Not Only SQL) databases are non-tabular databases that store data differently than relational tables. They are designed for large-scale data storage and high-performance requirements.

Feature	Relational (SQL)	NoSQL
Structure	Predefined Schema	Dynamic/Schema-less
Scaling	Vertical (Bigger server)	Horizontal (More servers)
Data Model	Tables with rows/columns	Documents, Key-Value, Graphs

2. Explain the types of NoSQL databases with examples.

- **Document Store:** Stores data in JSON/BSON format (e.g., **MongoDB**, CouchDB).
- **Key-Value Store:** Simple data storage using a unique key (e.g., **Redis**, DynamoDB).
- **Column-Family:** Optimized for reading large datasets (e.g., **Cassandra**, HBase).
- **Graph Database:** Stores nodes and relationships (e.g., **Neo4j**).

3. CAP Theorem in the context of NoSQL.

CAP Theorem states that a distributed system can only provide two out of three guarantees:

- **Consistency:** Every read receives the most recent write.
- **Availability:** Every request receives a response.
- **Partition Tolerance:** The system operates despite network failures.

Section B: Practical Task (MongoDB)

1. Insert 5 student records

```
db.students.insertMany([
  { "name": "Alice Smith", "age": 20, "course": "BSc IT",
    "skills": ["Java", "SQL"] },
  { "name": "Bob Jones", "age": 23, "course": "CS", "skills":
    ["Python", "C++"] },
  { "name": "Charlie Brown", "age": 21, "course": "BSc IT",
    "skills": ["HTML", "CSS"] },
  { "name": "David Wilson", "age": 22, "course": "Data
Science", "skills": ["R", "Python"] },
  { "name": "Eve Davis", "age": 24, "course": "CS", "skills":
    ["Go", "Docker"] }
]);
```

2. Find students in "BSc IT"

```
db.students.find({ "course": "BSc IT" });
```

3. Update a student's skills

```
db.students.updateOne(
  { "name": "Alice Smith" },
  { $push: { "skills": "MongoDB" } }
);
```

4. Delete a student record

```
db.students.deleteOne({ "name": "Bob Jones" });
```

5. Sort students by age

```
db.students.find().sort({ "age": 1 });
```

Section C: Case Study

Example: Amazon uses DynamoDB (NoSQL) for its shopping cart. During peak sales like Black Friday, millions of users add items simultaneously. A traditional SQL database might lock tables and crash under this load. NoSQL allows Amazon to scale horizontally across thousands of servers to ensure 100% availability.

Completed Assignment - Database Management Systems